Video Transcript - Part 17

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Hey everyone, welcome to part seventeen. In this video we're covering the confusion matrix, a simple table that shows exactly how a classification model is right and wrong. Despite the name, it's not confusing. Once you can read it, you can calculate precision, recall, and most of the metrics we've talked about.

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Here's the plan. We'll start with what a confusion matrix is and its four outcomes. Then we'll work through a numeric example, calculate the key metrics from it, and revisit the accuracy trap. After that, we'll see how to decide which error matters more, how thresholds change the results, and what to know about multi class problems and SageMaker.

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So what is a confusion matrix? It's a table that compares what your model predicted with what actually happened. For a yes or no problem, it's a two by two grid. Rows show the actual class, columns show the predicted class. One note: some tools flip the layout, so always check the labels. The four cells have names: true positive, true negative, false positive, and false negative. Here's a trick to read them. The second word is what the model predicted, positive or negative. The first word tells you whether that prediction was right, true or false.

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Let's make that concrete with fraud detection, where fraud is the positive class. A true positive is when the model flags a transaction as fraud and it really is fraud, a successful catch. A true negative is when the model says legitimate and it is legitimate, which is the normal, quiet case. A false positive is when the model flags a legitimate transaction as fraud. That's a false alarm, also called a Type one error. A false negative is when the model lets a real fraud through. That's a miss, also called a Type two error.

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Now some numbers. Imagine one thousand transactions, where ten are actually fraud and nine hundred ninety are legitimate. Our model catches eight of the ten frauds, those are true positives, and misses two, the false negatives. Of the legitimate transactions, it correctly passes nine hundred eighty six, the true negatives, but wrongly flags four, the false positives. So the model flagged twelve transactions in total, and eight of them were real fraud.

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From those four numbers we can calculate every key metric. Accuracy is true positives plus true negatives, divided by the total. That's eight plus nine eighty six, over a thousand, which is ninety nine point four percent. Precision is true positives divided by everything the model flagged. Eight out of twelve, about sixty seven percent. It answers, when the model says fraud, how often is it right? Recall, also called sensitivity, is true positives divided by all actual positives. Eight out of ten, or eighty percent. It answers, of all the real fraud, how much did we catch? Specificity is the same idea for negatives, nine eighty six out of nine ninety, about ninety nine point six percent. And the F one score balances precision and recall, which comes to about seventy three percent.

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Notice something. Accuracy is ninety nine point four percent, which sounds fantastic, but precision is only sixty seven percent and we missed twenty percent of the fraud. Now consider a lazy model that predicts not fraud every single time. It would have ninety nine percent accuracy, but a recall of zero, because it never catches anything. This is the accuracy trap we saw with imbalanced data, and the confusion matrix is how you spot it.

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So which metric should you focus on? It depends on which error costs more. If missing a real case is costly, like disease screening or fraud, optimize recall. If false alarms are costly, like a spam filter that hides real email, optimize precision. If you need a balance of both, use the F one score. And if your classes are balanced and both errors cost about the same, plain accuracy is fine.

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Most classifiers don't output a yes or no directly. They output a probability, and you choose a threshold to turn it into a decision. Lower the threshold and the model flags more cases, so recall goes up, but precision usually goes down because there are more false alarms. Raise the threshold and the opposite happens: fewer false alarms, but more misses. Each threshold gives you a different confusion matrix. The ROC curve plots the true positive rate against the false positive rate across all thresholds, and AUC summarizes it in a single number.

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A few more notes. With more than two classes, the confusion matrix simply grows to an N by N grid, and the diagonal shows the correct predictions. Look at the off diagonal cells to see which classes the model confuses with each other. When your classes are imbalanced, it helps to normalize each row into percentages, so you can compare fairly. And in SageMaker, you'll see these metrics in places like Autopilot model reports and Model Monitor's model quality checks.

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So to recap, the confusion matrix compares predictions with reality in four cells: true positives, true negatives, false positives, and false negatives. From it you calculate accuracy, precision, recall, specificity, and F one. Pick the metric that matches the cost of your errors, and never trust accuracy alone on imbalanced data. Thanks for watching, and I'll see you in the next one.